

2.3 Navigating the File System Concepts: Files, Directories, Paths, Home Directory, Parent-Child Relationship, Handling Command.- pwd, cd, mkdir, rmdir,ls Ordinary Files --_, handling commands: cat, cp,rm,mv, - cmp,comm,diff

File Naming Conventions:

In Unix, filenames:

- Maybe up to 255 characters long (in Linux8.0) –Some characters may require special handling.
- Can contain digits, a dot(.), a hyphen(-), or the underscore symbol(_) anywhere.
- Can contain both upper case as well as lower case alphabets.
- Are case sensitive.
e.g., RESUME.TXT is not same as resume.txt.
file.txt is not as same as File.txt or File.Txt.
- Should not have a blank, or a tab.

By rule, Unix file names do not have to have ending extensions (such as .txt or .exe) as do some other operating systems.

However, certain applications with which you interact may require extensions, such as Adobe's Acrobat Reader (.pdf) or a web browser (.html). Thus the following are all valid Unix file names (note these may be any file type):

My_Stuff	a file named My_Stuff
my_stuff	a different file than above
mortgage.c	a C language program...can you guess what it does?
a.out	a C language binary executable
.profile	ksh default startup file

The Directory:

A directory is a file the solo job of which is to store the file names and the related information. All the files, whether ordinary, special, or directory, are contained in directories.

Unix uses a hierarchical structure for organizing files and directories. This structure is often referred to as a directory tree. The tree has a single root node, the slash character (/), and all other directories are contained below it.

Home Directory

The directory in which you find yourself when you first login is called your home directory.

You will be doing much of your work in your home directory and subdirectories that you'll be creating to organize your files.

You can go in your home directory anytime using the following command –

```
$cd ~
```

```
$
```

Here ~ indicates the home directory. Suppose you have to go in any other user's home directory, use the following command –

```
$cd ~username
```

```
$
```

The Home Directory:

The system places a user in his HOME directory upon login.

- Every user has a home directory, typically under/home or under/user
- Contains user- specific configuration files as well as user data.
- Represented by the ~ character.

Current Working Directory:

When you are in a shell, you are always in a directory on the system, and that directory is called the *current working directory*. When you first log in to the system, your home directory is the current working directory.

A container which stores information is known as File. You can treat it simply as a sequence of characters. It can be a document, picture, audio or video stream, data library, program file, application, or other collection of data.

Very important point here, Linux treats directories and devices as a file. In Linux system a directory is a folder which keeps filenames and other sub-directories. A Linux system considers all physical devices like your hard drive, CD-ROM, printer, memory etc. as files.

Imp: In Linux system a shell is also a file and kernel too.

Linux File Types

In UNIX there are different types of categories are there.

1. Ordinary File or Regular File
2. Directory File
3. Device File
4. Symbolic link file or soft link file
5. FIFO (Named Pipe)
6. Socket

1. Ordinary File

An ordinary file also known as regular file commonly is the most common file type in Linux system. All programs (C, C+, Java etc.) belongs to this file type.

Ordinary file is divided into two types such as Text File and Binary File.

Text File:: It contains only printable ASCII character within the range 0-127. Often we can view the content of a text file and can make sense out of it. Here every line ends with a new line character, created by ENTER Key.

For Example: All 'C', Source Codes, all Python Scripts, Shell Scripts

Binary File:: It contains both printable as well as non-printable characters within the ASCII range 0-255. We can not make any sense after viewing a binary file because it contains ASCII symbols.

For Example: All C and Java Object codes, all image, video, audio files are binaries. And also most Linux/UNIX commands are Binary File type.

When displaying the Binary File type with cat command it produces unreadable output and it may also disturb your terminal settings.

2. Directory File

Very important point you should keep in your mind that some users say and sometimes it is commonly said that a directory contains files and other directories. But it contains their names and inode number of a file or directory.

If you have directory and 10 files in a directory, there will be only 10 entries resides in the directory. It contains 2 attributes of a file.

- The File name
- Inode Number(A unique identification number of every newly created file which system assigns)

Unlike windows folders Linux directories never contain file data, file data are stored in data blocks.

Tip: *A directory contains filename and not the file's contents.*

3. Device File

Linux or UNIX considers every hardware devices as a file. They have different types of names also. All these device files are stored in the directory (/dev). A device file is indeed special it's not really a stream of characters. In fact it does not contain anything at all.

4. Symbolic link File

Symbolic link file also known as Soft link file. It is a very special type of file which contains path of the original file. Also you can say a symbolic link is a reference to another file.

5. FIFO (Named Pipe)

One of the main advantage and strength of Linux or Unix it has always been inter-process communication. Among the facilities provided by the OS are *pipes*, which connect the output of one process to the input of another. This is fine if both processes exist in the same parent process space, started by the same user, but there are circumstances where the communicating processes must use FIFOs, here referred to as *named pipes*. One such circumstance occurs when the processes must be executed under different user names and permissions.

Named pipes are special files that can exist anywhere in the file system. They can be created with the command `mkfifo`

6. Socket

A Unix/Linux socket is a special file which allows for advanced inter-process communication. A Unix Socket is used in a client-server application framework. In essence, it is a stream of data, very similar to network stream (and network sockets), but all the transactions are local to the filesystem. Unlike named pipes which allow only unidirectional data flow, sockets are fully duplex-capable.

7. Hidden Files

An invisible file is one, the first character of which is the dot or the period character (.). Unix programs (including the shell) use most of these files to store configuration information.

Some common examples of the hidden files include the files –

- `.profile` – The Bourne shell (`sh`) initialization script
- `.kshrc` – The Korn shell (`ksh`) initialization script
- `.cshrc` – The C shell (`csh`) initialization script
- `.rhosts` – The remote shell configuration file

To list the invisible files, specify the -a option to ls –

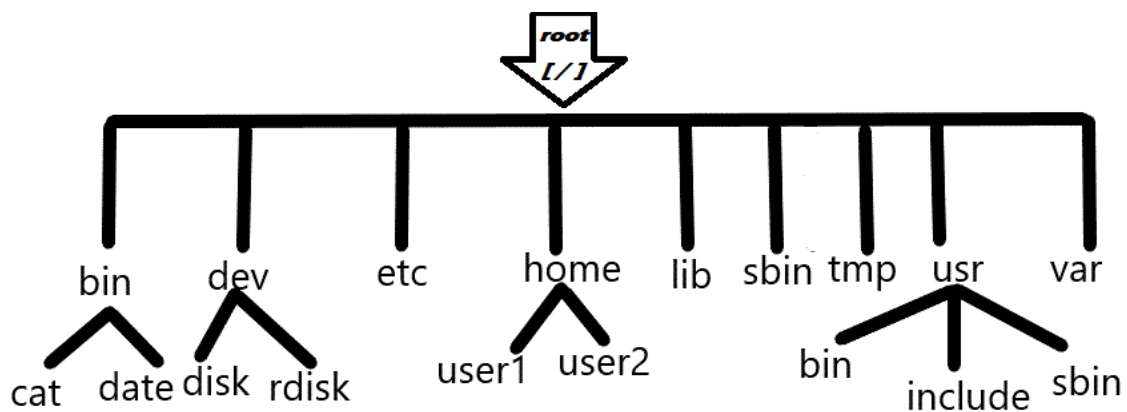
```
$ ls -a
```

```
.          .profile      docs      lib      test_results
..         .rhosts      hosts     pub      users
.emacs     bin          hw1       res.01   work
.exrc      ch07          hw2       res.02
.kshrc     ch07.bak      hw3       res.03
$
```

- Single dot (.) – This represents the current directory.
- Double dot (..) – This represents the parent directory.

Parent Child Relationship

All the files in Linux system are related to each other. The collection of all these files (Ordinary, directory, device etc.) is called as file system. And the filesystem in Linux is organized in a hierarchical structure. Directories/Files are here in a parent child relationship.



Linux file system is considered as single monolithic tree, an inverted one, root upwards. As you can see this tree is organized in hierarchical structure.

The directories and its contains.

Path	Type of Data Holds
/bin	Stands for binary. It contains command files of maximum commands
/dev	It contains all device files
/etc	This directory contains system configuration.
/home	Home directory contain all users
/lib	This directory contains all C language library files
/sbin	Super user bin. This dir contains command files of administrative commands
/stand/unix	this is unix kernel
/boot/vmlinuz	Path of Linux kernel path
/tmp	It contains temporary files, system wipes away this files in regular interval
/usr or /usr/bin	this directory contains command files of few additional commands
/usr/include	This dir. contain all C language header files
/var	This is variable part of the file system. Contains all user mailbox, crontab file of all user
/var/log	Contains system logs

Path of A file

A file path is the human-readable representation of a file or directory location on a computer system. The location of a file or a directory can be specified by either of two methods : by its *absolute* pathname or its *relative* pathname.

Absolute Pathnames:

- Absolute pathnames begin with a slash(/)
- It's a complete path to the file location
- It can be used any time you wish to specify a filename

Relative Pathname:

- Relative pathname do not begin with /.
- Specifies location relative to your current working directory.
- Can be used as a shorter way to specify filename.

Wildcard Characters:

Unix offers the facility to perform an operation on the set of files without having to specifying all the file names on the operation has to be performed. This is made possible by use of certain special characters in the command in place of the actual file names. These special characters are interpreted by Unix as a specific pattern of characters. Unix then compares all the filenames under the directory specified in the command to find out the filenames that match the pattern. The command is executed on the files with names that match the pattern.

Table1:

Character	Purpose
*	Matches none or one character or a string of more than one character
?	Matches exactly one character
[]	Matches exactly one of a specified set of characters

Refer following link for more information

http://dsl.org/cookbook/cookbook_8.html

https://homepages.uc.edu/~thomam/Intro_Unix_Text/File_System.html

Handling Command

1. pwd (print working directory)

This command displays the absolute path of the current working directory.

Execute the following command and write the output:

```
$pwd
```

2. cd (change directory)

The following command changes to/usr directory.

```
$cd/usr
```

Note: When *cd* command is executed without specifying directory name, then it will switch to the user's HOME directory.

```
$cd
```

3. mkdir(make directory)

If you are logged is as pramod:

The following command creates a subdirectory *dcn* under current directory. However , the new directory doesn't become the current directory.

```
$ mkdir dcn
```

Completes path names can also be specified with *mkdir* as:

```
$ mkdir/usr/pramod/dcn
```

4. rmdir(remove directory)

The following command removes the *dcn* subdirectory from the current directory.

```
$rmdir dcn
```

Note: A directory can be deleted only if it is:

- Empty(does not contains files or subdirectories)
- Not the current directory

Complete path names can also be specified with *mkdir* as shown in *mkdir command*.

5. ls (list)

Following command displays the file under current directory.

```
$ ls
abc.txt
dcn
$
```

Table1: Options for ls command:

Option	Functions
-a	Displays all files including.,...and hidden files
-A	Displays all but not. and ..
-C	List entries by column
-d	List directory entries instead of contents
-l	Use long listing format
-r	Sort in reverse order
-R	List subdirectories recursively

-s	Prints size of each file in blocks
-S	Sort by file size
-t	Sort by modification time,-u sort by access time
V	Sort by version
1	List one file per line
X	List in column format with row wise sorting
F	Classifies files, executables, directories
i	Shows icode no. of a file

Ordinary Files handling commands:

1. cat (concatenate)

It concatenates the contents of a specified file and also useful for creating and displaying contents of a file.

- To display the contents of a file *info.txt*,
\$ cat *info.txt*
- To create a file
\$ cat> abc.txt
Unix is a multi-programming operating system.
Unix is a robust and powerful operating system.
One should learn to use it.
<ctrl-d>
\$
Ctrl-d is used to save file and exit cat command.
Execute the following options and write the use of it.

2. cp(copy)

The following command copies *resume.txt* file to *newresume.txt*.

```
$cp resume.txt newresume.txt
```

The following command will copy all the files which start with *ch* and ends with exactly two character having extension txt from current directory to */usr/pramod/dcn* directory.

```
$ cp ch??txt /usr/pramod/dcn
```

Execute the following options and write the use of it.

3. mv (move)

The following command moves(renames) *resume.txt* to *biodata.txt*.

```
$ mv resume.txt biodata.txt
```

The following command moves *resume.txt* file to */usr/pramod* directory

```
$ mv resume.txt to /usr/pramod
```

The following command moves all the files with extension .sh from the current directory to */usr/pramod/dcn* directory

```
$ mv *.sh /usr/pramod/dcn
```

Option -i as available which behaves exactly like in cp.

4. The cmp command

The cmp command is used to compare files. When used without options, it uses two file names as arguments and displays the differences on the terminal.

Syntax: `cmp file1 file2`

The option `-l` gives a detailed list of the byte number and the differing byte in octal for each character that differs in both files.

5. The comm command

The comm command compares two sorted files. It compares each line of the first file with its corresponding line in the second file. It displays output in three columns: lines unique to the first file, lines unique to the second file, and lines common to both files, respectively.

Syntax: `comm file1 file2`

6. The diff command

To show differences between two text files, it tells you which lines in one file have to be changed to make the two files identical. Diff uses certain special symbols and instructions to indicate the changes that are required to make two files identical.

Syntax: `diff file1 file2`